Data Cleaning

```r
# ============================================================
# 02_data_cleaning.R
# Airline Service Performance Visual Analytics
# ============================================================


# 1. LOAD PACKAGES -------------------------------------------------------

library(readxl)
library(dplyr)


# 2. FILE PATHS ----------------------------------------------------------

file_path <- "D:/data-portfolio/08_airline_service_performance_visual_analytics/data/raw/Dataset Assignm

output_path <- "D:/data-portfolio/08_airline_service_performance_visual_analytics/data/processed/airlin


# Make sure the processed folder exists
dir.create(
  dirname(output_path),
  recursive = TRUE,
  showWarnings = FALSE
)


# 3. DEFINE CLEAN COLUMN NAMES --------------------------------------------

clean_column_names <- c(
  "Route",
  "Departing_Port",
  "Arriving_Port",
  "Airline",
  "Month",
  "Sectors_Scheduled",
  "Sectors_Flown",
  "Cancellations",
  "Departures_On_Time",
  "Arrivals_On_Time",
  "Departures_Delayed",
  "Arrivals_Delayed",
  "OnTime_Departures_Percent",
  "OnTime_Arrivals_Percent",
  "Cancellations_Percent"
)
```

```r
# 4. CHECK AVAILABLE SHEETS -----------------------------------------------

sheet_names <- excel_sheets(file_path)

cat("\n=== ORIGINAL SHEETS ===\n")
```

```
##
## === ORIGINAL SHEETS ===
```

```r
print(sheet_names)
```

```
##  [1] "2020-23 OTP" "2020"        "2019"        "2018"        "2017"
##  [6] "2016"        "2015"        "2014"        "2013"        "2012"
## [11] "2011"        "2010"
```

```r
# 5. EXCLUDE THE DUPLICATED 2020 SHEET ------------------------------------

sheets_to_use <- sheet_names[
  sheet_names != "2020"
]

cat("\n=== SHEETS USED FOR CLEAN DATA ===\n")
```

```
##
## === SHEETS USED FOR CLEAN DATA ===
```

```r
print(sheets_to_use)
```

```
##  [1] "2020-23 OTP" "2019"        "2018"        "2017"        "2016"
##  [6] "2015"        "2014"        "2013"        "2012"        "2011"
## [11] "2010"
```

```r
# 6. READ AND COMBINE THE SHEETS ------------------------------------------

data_list <- list()

for (sheet in sheets_to_use) {

  data <- read_excel(
    file_path,
    sheet = sheet,
    range = cell_cols("A:O"),
    na = c("", "na", "NA")
  )

  names(data) <- clean_column_names

  data$Source_Sheet <- sheet
```

```r
  data_list[[sheet]] <- data
}

master_raw <- bind_rows(data_list)


cat("\n=== RAW COMBINED DATA ===\n")
```

```
##
## === RAW COMBINED DATA ===
```

```r
cat(
  "Rows before cleaning:",
  nrow(master_raw),
  "\n"
)
```

```
## Rows before cleaning: 80977
```

```r
# 7. REMOVE NON-DATA ROWS -------------------------------------------------

clean_data <- master_raw %>%
  filter(
    !is.na(Route),
    Route != "",
    !is.na(Month)
  )


cat("\n=== NON-DATA ROW CHECK ===\n")
```

```
##
## === NON-DATA ROW CHECK ===
```

```r
cat(
  "Rows removed:",
  nrow(master_raw) - nrow(clean_data),
  "\n"
)
```

```
## Rows removed: 5
```

```r
cat(
  "Rows remaining:",
  nrow(clean_data),
  "\n"
)
```

```
## Rows remaining: 80972
```

```r
# 8. CLEAN TEXT COLUMNS ---------------------------------------------------

clean_data$Route <- trimws(
  clean_data$Route
)

clean_data$Departing_Port <- trimws(
  clean_data$Departing_Port
)

clean_data$Arriving_Port <- trimws(
  clean_data$Arriving_Port
)

clean_data$Airline <- trimws(
  clean_data$Airline
)


# 9. CLEAN THE DATE COLUMN ------------------------------------------------

clean_data$Month <- as.Date(
  clean_data$Month
)


# 10. CONVERT OPERATIONAL COLUMNS TO NUMERIC ------------------------------

numeric_columns <- c(
  "Sectors_Scheduled",
  "Sectors_Flown",
  "Cancellations",
  "Departures_On_Time",
  "Arrivals_On_Time",
  "Departures_Delayed",
  "Arrivals_Delayed",
  "OnTime_Departures_Percent",
  "OnTime_Arrivals_Percent",
  "Cancellations_Percent"
)

clean_data[numeric_columns] <- lapply(
  clean_data[numeric_columns],
  as.numeric
)


# 11. KEEP ORIGINAL AIRLINE LABEL -----------------------------------------

clean_data$Airline_Original <- clean_data$Airline


# 12. STANDARDISE AIRLINE NAMES -------------------------------------------
```

```r
clean_data$Airline <- case_when(

  clean_data$Airline == "virgin Australia" ~
    "Virgin Australia",

  clean_data$Airline == "Regional Express" ~
    "Rex Airlines",

  TRUE ~ clean_data$Airline
)


cat("\n=== AIRLINE NAME CLEANING ===\n")
```

```
##
## === AIRLINE NAME CLEANING ===
```

```r
cat(
  "Original airline labels:",
  n_distinct(clean_data$Airline_Original),
  "\n"
)
```

```
## Original airline labels: 14
```

```r
cat(
  "Clean airline labels:",
  n_distinct(clean_data$Airline),
  "\n"
)
```

```
## Clean airline labels: 12
```

```r
cat("\nOriginal airline labels:\n")
```

```
##
## Original airline labels:
```

```r
print(
  sort(
    unique(clean_data$Airline_Original)
  )
)
```

```
##  [1] "All Airlines"
##  [2] "Bonza"
##  [3] "Jetstar"
##  [4] "Qantas"
##  [5] "QantasLink"
##  [6] "Regional Express"
```

```
##  [7] "Rex Airlines"
##  [8] "Skytrans"
##  [9] "Skywest"
## [10] "Tigerair Australia"
## [11] "virgin Australia"
## [12] "Virgin Australia"
## [13] "Virgin Australia - ATR/F100 Operations"
## [14] "Virgin Australia Regional Airlines"
```

```r
cat("\nClean airline labels:\n")
```

```
##
## Clean airline labels:
```

```r
print(
  sort(
    unique(clean_data$Airline)
  )
)
```

```
##  [1] "All Airlines"
##  [2] "Bonza"
##  [3] "Jetstar"
##  [4] "Qantas"
##  [5] "QantasLink"
##  [6] "Rex Airlines"
##  [7] "Skytrans"
##  [8] "Skywest"
##  [9] "Tigerair Australia"
## [10] "Virgin Australia"
## [11] "Virgin Australia - ATR/F100 Operations"
## [12] "Virgin Australia Regional Airlines"
```

```r
# 13. CHECK MISSING OTP VALUES --------------------------------------------

missing_otp <- clean_data %>%
  filter(
    is.na(OnTime_Departures_Percent) |
    is.na(OnTime_Arrivals_Percent)
  )


cat("\n=== MISSING OTP CHECK ===\n")
```

```
##
## === MISSING OTP CHECK ===
```

```r
cat(
  "Rows with missing OTP:",
  nrow(missing_otp),
  "\n"
)
```

```
## Rows with missing OTP: 91
```

```r
cat(
  "Missing Departure OTP values:",
  sum(
    is.na(
      clean_data$OnTime_Departures_Percent
    )
  ),
  "\n"
)
```

```
## Missing Departure OTP values: 91
```

```r
cat(
  "Missing Arrival OTP values:",
  sum(
    is.na(
      clean_data$OnTime_Arrivals_Percent
    )
  ),
  "\n"
)
```

```
## Missing Arrival OTP values: 91
```

```r
# 14. CHECK ZERO SECTORS FLOWN ------------------------------------------

zero_flown <- clean_data %>%
  filter(
    Sectors_Flown == 0
  )


cat("\n=== ZERO SECTORS FLOWN CHECK ===\n")
```

```
##
## === ZERO SECTORS FLOWN CHECK ===
```

```r
cat(
  "Rows with zero sectors flown:",
  nrow(zero_flown),
  "\n"
)
```

```
## Rows with zero sectors flown: 91
```

```r
# 15. VALIDATE MISSING OTP VALUES ----------------------------------------

all_missing_have_zero_flown <- all(
  missing_otp$Sectors_Flown == 0
```

```r
)

all_zero_flown_have_missing_otp <- all(
  is.na(
    zero_flown$OnTime_Departures_Percent
  ) &
  is.na(
    zero_flown$OnTime_Arrivals_Percent
  )
)


cat("\n=== OTP VALIDATION ===\n")
```

```
##
## === OTP VALIDATION ===
```

```r
cat(
  "All missing OTP rows have zero sectors flown:",
  all_missing_have_zero_flown,
  "\n"
)
```

```
## All missing OTP rows have zero sectors flown: TRUE
```

```r
cat(
  "All zero-sector rows have missing OTP:",
  all_zero_flown_have_missing_otp,
  "\n"
)
```

```
## All zero-sector rows have missing OTP: TRUE
```

```r
# Missing OTP values are kept as NA.
# OTP is undefined when no sectors were flown.


# 16. CHECK DUPLICATED ANALYTICAL KEYS ------------------------------------

duplicate_keys <- clean_data %>%
  count(
    Route,
    Departing_Port,
    Arriving_Port,
    Airline,
    Month
  ) %>%
  filter(
    n > 1
  )


cat("\n=== DUPLICATE KEY CHECK ===\n")
```

```
##
## === DUPLICATE KEY CHECK ===
```

```
cat(
  "Duplicated analytical keys:",
  nrow(duplicate_keys),
  "\n"
)
```

```
## Duplicated analytical keys: 0
```

```
print(
  head(
    duplicate_keys,
    10
  )
)
```

```
## # A tibble: 0 x 6
## # i 6 variables: Route <chr>, Departing_Port <chr>, Arriving_Port <chr>,
## #   Airline <chr>, Month <date>, n <int>
```

```
# 17. VALIDATE FLIGHT COUNT RELATIONSHIPS ---------------------------------

# A small tolerance avoids false errors caused by
# tiny floating-point differences.

tolerance <- 0.000001


scheduled_check <- clean_data %>%
  filter(
    abs(
      Sectors_Scheduled -
      Sectors_Flown -
      Cancellations
    ) > tolerance
  )


departure_check <- clean_data %>%
  filter(
    abs(
      Sectors_Flown -
      Departures_On_Time -
      Departures_Delayed
    ) > tolerance
  )


arrival_check <- clean_data %>%
  filter(
```

```r
    abs(
      Sectors_Flown -
      Arrivals_On_Time -
      Arrivals_Delayed
    ) > tolerance
  )


cat("\n=== FLIGHT COUNT VALIDATION ===\n")
```

```
##
## === FLIGHT COUNT VALIDATION ===
```

```r
cat(
  "Scheduled count inconsistencies:",
  nrow(scheduled_check),
  "\n"
)
```

```
## Scheduled count inconsistencies: 0
```

```r
cat(
  "Departure count inconsistencies:",
  nrow(departure_check),
  "\n"
)
```

```
## Departure count inconsistencies: 0
```

```r
cat(
  "Arrival count inconsistencies:",
  nrow(arrival_check),
  "\n"
)
```

```
## Arrival count inconsistencies: 0
```

```r
# 18. FINAL DATASET CHECK -------------------------------------------------

cat("\n=== FINAL CLEAN DATASET ===\n")
```

```
##
## === FINAL CLEAN DATASET ===
```

```r
cat(
  "Number of rows:",
  nrow(clean_data),
  "\n"
)
```

```
## Number of rows: 80972
```

```r
cat(
  "Number of columns:",
  ncol(clean_data),
  "\n"
)
```

```
## Number of columns: 17
```

```r
cat(
  "Date range:",
  as.character(
    min(clean_data$Month)
  ),
  "to",
  as.character(
    max(clean_data$Month)
  ),
  "\n"
)
```

```
## Date range: 2010-01-01 to 2024-02-01
```

```r
cat(
  "Number of months:",
  n_distinct(clean_data$Month),
  "\n"
)
```

```
## Number of months: 170
```

```r
airport_names <- unique(
  c(
    clean_data$Departing_Port,
    clean_data$Arriving_Port
  )
)

airport_names <- airport_names[
  !is.na(airport_names) &
  airport_names != "All Ports"
]


cat(
  "Number of physical airports:",
  length(airport_names),
  "\n"
)
```

```
## Number of physical airports: 42
```

```r
cat(
  "Number of clean airline labels:",
  n_distinct(clean_data$Airline),
  "\n"
)
```

```
## Number of clean airline labels: 12
```

```r
cat(
  "Number of route labels:",
  n_distinct(clean_data$Route),
  "\n"
)
```

```
## Number of route labels: 155
```

```r
# 19. SAVE CLEAN DATA -----------------------------------------------------

write.csv(
  clean_data,
  output_path,
  row.names = FALSE,
  na = ""
)


cat("\n=== FILE EXPORTED ===\n")
```

```
##
## === FILE EXPORTED ===
```

```r
cat(
  "Clean data saved to:",
  output_path,
  "\n"
)
```

```
## Clean data saved to: D:/data-portfolio/08_airline_service_performance_visual_analytics/data/processed
```

```r
# 20. CLEANING FINISHED ---------------------------------------------------

cat("\n=== DATA CLEANING FINISHED ===\n")
```

```
##
## === DATA CLEANING FINISHED ===
```